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MILITARY SPECIFICATION

DYE, 1, 4-DIMETHYLAMINO-ANTHRAQUINONE (For Colored Smoke Mixtures)

1. SCOPE

1.1 This specification covers, 1, 4-Dimethylamino-anthraquinone (blue dye) for use in the manufacture of colored smoke mixtures.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Federal

PPP-D-723	Drums, Fiber
PPP-D-729	Drums, Metal, 55 Gallon (for Shipment of Non-Corrosive Materials)
RR-S-366	Sieves, Standard, for Testing Purposes

Military

MIL-P-116	Preservation, Methods of

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PSC 1375

STANDARDS

Military

MIL-STD-129 Marking for Shipment and Storage

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Ratings, Rules and Regulations

(Application for copies of the above publication should be addressed to the Uniform Classification Committee, 202 Chicago Union Station, Chicago 6, Illinois)

AMERICAN TRUCKING ASSOCIATION

National Motor Freight Classification Rules on Container Specifications

(Application for copies of the above publication should be addressed to the American Trucking Association, 1424 16th St., NW, Washington, D.C.)

BECKMAN BULLETIN 89-E

Instructions for the Ultraviolet Accessory Set

(Application for copies of the above publication should be addressed to the Beckman Instruments, Inc.)

3. REQUIREMENTS

3.1 Volatile matter. The volatile matter content of the material shall be 0.5 percent, maximum (max.), when determined as specified in 4.4.1.

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3.2 Purity. The purity of the material shall be 84.0 percent, minimum (min.), when determined as specified in 4.4.2.

3.3 Melting point. The melting point of the material shall be 209 degrees centigrade (°C), min., when determined as specified in 4.4.3.

3.4 Sodium chloride content. The sodium chloride content of the material shall be 5.0 percent, max., when determined as specified in 4.4.4.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified herein the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Initial production lot. The initial production lot shall be the first lot of a series of production lots as defined in 4.2.2. In addition, should any change in chemical process or physical mixing process or production equipment occur at any time during the processing phase of the dye, the first lot produced thereafter under the altered conditions shall be considered to be a new initial production lot and shall be treated thereas.

4.2.1 Sampling for initial production lot. The contractor shall submit one 2 ounce sample of the dye to a Government Approved Laboratory for testing as specified in 4.4.

4.2.2 Production lot. A production lot shall consist of dye from not more than one manufacturing batch. A batch is defined as that quantity of dye which has been subjected to the same unit chemical process or physical mixing process intended to make the final product substantially uniform.

4.3 Inspection

4.3.1 Certification (applicable to all production lots except the initial production lot). The contractor shall certify that the 1, 4-Dimethylamino-Anthraquinone complies with the applicable

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requirements for the following determinations:

a. Volatile matter
b. Purity
c. Melting point
d. Sodium chloride

The certificate furnished to the Government inspector, prior to or at the time of delivery showing tests and inspections conducted and results obtained, shall be signed by a responsible agent of the certifying organization and shall be accompanied by evidence of the agent's authority to bind his principal. The Government reserves the right to check test material submitted by the contractor on certification.

4.3.2 Packing and marking. It shall be ascertained that the packing of the dye and the marking of the container conform to this specification.

4.4 Test procedures

4.4.1 Determination of volatile matter. An accurately weighed portion of approximately 5 grams (gm.) of the dye shall be transferred to a tared, glass-stoppered weighing bottle, 50 to 75 millimeter (mm.) in diameter by 25 to 30 mm. in height. The bottle and contents shall be accurately weighed and dried for 2 hours at 80° ± 2°C., then cooled to room temperature in a desiccator and reweighed. The percentage of volatile matter shall be calculated from the loss in weight. The dried sample shall be retained for the following determination.

4.4.2 Determination of purity. An accurately weighed portion of approximately 1.00 gm. of the dye previously dried for 2 hours at 80° ± 2°C shall be transferred to a 600 milliliter (ml.) beaker and about 250 ml. of 95 percent ethyl alcohol shall be added. The contents of the beaker shall be heated on a steam bath until the soluble material of the sample has dissolved, and the supernatent liquid decanted into a 1-liter volumetric flask. An additional 100 ml. of alcohol shall be added to the residue contained in the beaker, and the beaker heated and the solution decanted as before. This procedure shall be repeated until the supernatent liquid is colorless. The contents of the flask shall be diluted to the mark with 95 percent alcohol, the temperature of the solution shall be adjusted to room temperature, and again diluted to the mark (solution "A"). The contents of the flask shall be mixed thoroughly. Using a pipette, a 10 ml. aliquot of solution

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"A" shall be transferred to a 100 ml. volumetric flask and diluted to the mark with alcohol as before. (Solution "B"). In the same manner, a 10 ml. aliquot of solution "B" shall be diluted to 100 ml. with alcohol (solution "C"). The absorbancy of solution "C" at wavelengths of 410 millimicron (mu.) and 640 mu. shall be determined using a blank of 95 percent ethyl alcohol taken from the same lot that was used in the preparation of the sample solutions. A wavelength calibration should be made against suitable spectral lines, using a mercury source at 640 mu. region, and a hydrogen source at the alpha hydrogen line of 656.3 mu. (see 6.2). The percent of 1, 4-dimethylamino-anthraquinone shall be calculated as follows:

$$\text{Percent 1, 4 dimethylamino-anthraquinone} = \frac{A640}{65.2 \times b \times c} \times 100$$

Where:
A640 = absorbancy of the solution at 640 mu.
b = cell depth, cm.
c = concentration of sample gm. per liter = 0.01 times weight of sample
65.2 = absorbancy index of 1, 4 dimethylamino-anthraquinone.

4.4.3 Determination of melting point.

4.4.3.1 Apparatus. A melting point bath shall be set up equipped with a mechanical stirrer and a source of heat that can be easily regulated. A beaker of 1 or 2 liters capacity about three-fourths full of clean paraffin oil is recommended. An accurately standardized thermometer shall be suspended in the bath so that the bulb is not less than 1 1/2 inches from the bottom of the bath; if the mercury column will not be completely immersed at the temperature of the observed melting point, a second thermometer shall be suspended about 1/2 inch from the first thermometer with its bulb about the height of the middle of the exposed mercury column of the first thermometer.

4.4.3.2 Procedure. The dried dye retained from the determination of volatile matter shall be used, or a small portion (not over 5 gm.) of the dye shall be ground so that it will pass a No. 100 U. S. Standard sieve complying with Specification RR-S-366. It shall be dried for 2 hours at 100° to 105°C, or over sulfuric acid in a desiccator for 24 hours. Thin walled capillary tubes of uniform diameter, long enough to extend to the top

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of the bath shall be used. The tube shall be filled with ground dye to a depth of approximately 4 mm. and compacted by tapping, the tube shall be fastened to the standardized thermometer so that the lower end of the tube is in contact with the bulb of the thermometer. The stirrer shall be started, and the bath heated rapidly to about 200°C., and then gradually so that the rise in temperature does not exceed 1°C in 3 minutes. The temperature shall be recorded when observed at the instant the meniscus of melted dye first appears across the capillary tube.

4.4.3.3 Correction. If the mercury column is completely immersed at the temperature at which the meniscus of melted dye first appears, this temperature shall be reported as the melting point of the sample. If part of the column is exposed, the following correction shall be added to the observed melting point.

$$n\ (T - t) \times 0.000159$$

Where:
n = number of degrees in the exposed column.
T = uncorrected melting point.
t = average temperature of exposed column, that is, temperature indicated by second thermometer.
0.000159 = coefficient of expansion of mercury in glass.

4.4.4 Sodium chloride. An accurately weighed portion of approximately 0.3 gm of the dye previously dried for 2 hours at 80° $\pm$ 2°C shall be transferred to a 400 ml. beaker and moistened with a few ml. of 95 percent ethyl alcohol. Approximately 250 ml. of distilled water shall be added and the solution stirred for a few minutes to dissolve the sodium chloride. The suspension shall be filtered, using a No. 40 Whatman filter paper or equivalent. The residue on the filter paper shall be washed with five 10 ml. portions of distilled water directed from a wash bottle, adding the washings to the filtrate. Five ml. of concentrated nitric acid and 15 ml. of 0.1N silver nitrate ($AgNO_3$) solution shall be added to the filtrate. The suspension shall be heated almost to boiling, and allowed to stand in the dark for 4 hours. The supernatant liquid shall be tested for completeness of precipitation with a few drops of $AgNO_3$ solution, then the precipitate collected on a tared, medium porosity sintered glass crucible. The precipitate on the crucible shall be washed with 0.01N nitric acid solution until the washings show

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no turbidity when tested with one or two drops of 0.1 N hydrochloric acid solution. The precipitate shall be washed with several 10 ml. portions of acetone to dissolve out any occluded dye. The crucible shall be aspirated until the odor of acetone can no longer be detected. The crucible and contents shall be dried for 2 hours at 135°C to 140°C., cooled in a desiccator, and weighed. The sodium chloride content shall be calculated as follows:

$$\text{Percent sodium chloride} = \frac{40.75A}{W}$$

Where:
A = gain in weight of crucible in grams.
W = weight of sample in grams.

4.4.5 Rejection (applicable to initial production lot only). The initial production lot shall be rejected if the sample fails to comply with any of the requirements specified. In the event of rejection, the first subsequent lot processed thereafter shall be considered the initial production lot and subjected to the specified testing procedure. This policy shall be continued until such time that an initial production lot has been found which complies with the requirements.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Not applicable

5.2 Packing

5.2.1 Level A

5.2.1.1 Exterior containers. The dye shall be packed in accordance with Method IA-5 or IA-13 of Specification MIL-P-116 in 55 gal. steel drums with full removable heads in accordance with Specification PPP-D-729, Type III or in fiber drums in accordance with Specification PPP-D-723, Type III, Grade A. Each drum shall be lined with 0.004 inches (nominal) thick polyethylene and the liner shall be sealed by knotting, tying or heat sealing.

5.2.2 Level B. Not applicable

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5.2.3 Level C. The dye shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall supply with Uniform Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

5.3 Marking

5.3.1 Special markings. Exterior shipping containers shall be marked as follows: "CAUTION - AVOID CONTACT OR BREATHING OF DUST OR FUMES. PROVIDE ADEQUATE VENTILATION".

5.3.2 Normal markings. In addition to the markings required by contract or order, unit packages, intermediate packages (when used), and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Level of protection (see 6.1.1).
c. Government Approved Laboratory (see 4.2).

6.1.1 Criteria for use of proper level of packing shall be as follows:

For Level A. This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

For Level C. This level shall be used only when it is definitely known that the packed item is to be shipped to domestic installations for immediate use at the first receiving activity.

6.2 A calibrated Beckman quartz spectrophotometer, model DU having a cell depth of approximately 1.00 cm. was found to be satisfactory for this measurement. Instructions for the calibration of the Beckman model DU quartz spectrophotometer are contained in Beckman Bulletin 89-E, "Instructions for the Ultraviolet Accessory Set ", Beckman Instruments, Inc.

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